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This procedure is applicable to below models:

V810 STD	☐	S1	☐	S2
V810 XXL	☐	S1	☐	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	30 May 2016	First Release	Tan Liang Kai

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Belt Relay

Relay removal

1. Close X-ray Source.Open Front Door.
2. Power off SSCA

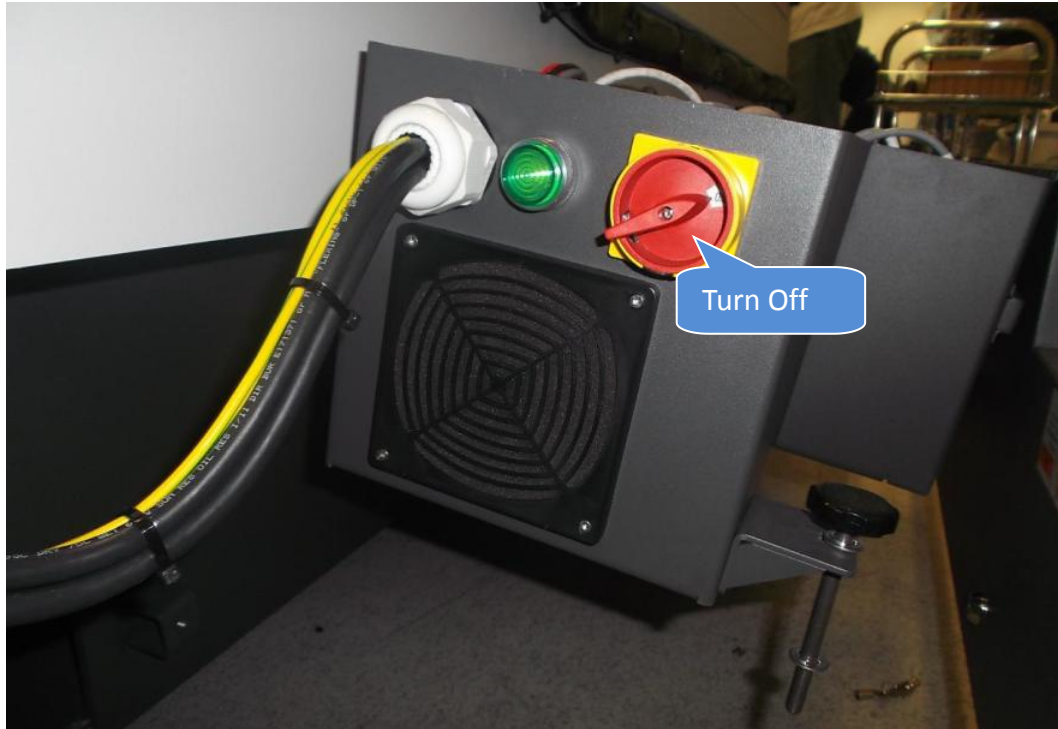


Figure 1 SSCA Power Off

3. Gently loosen Both Anchors as below Figure 2

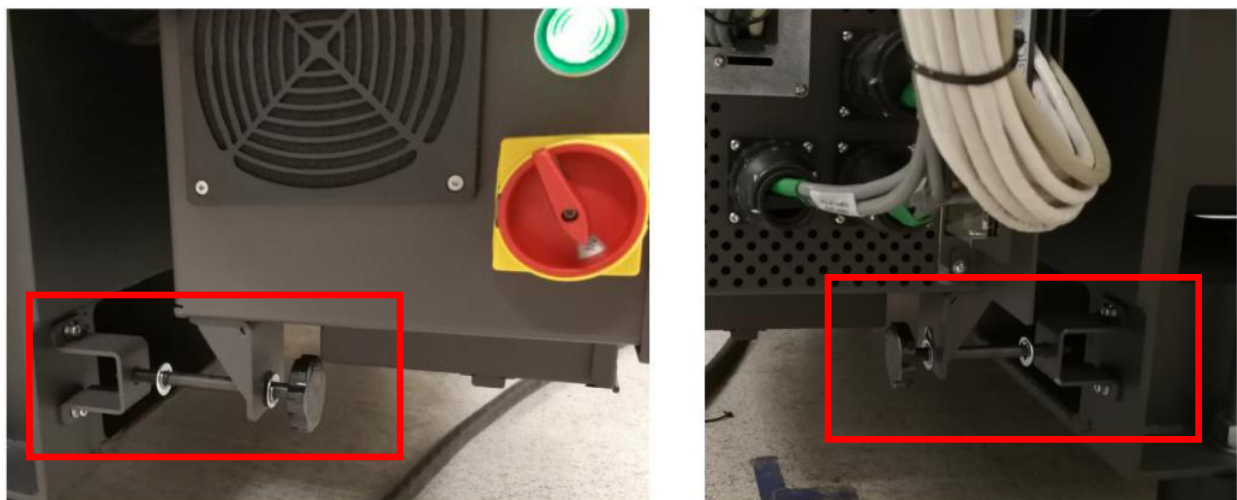


Figure 2 Anchors Release

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4. Extend SSCA Support Rod which locate underneath of SSCA as below Figure 4

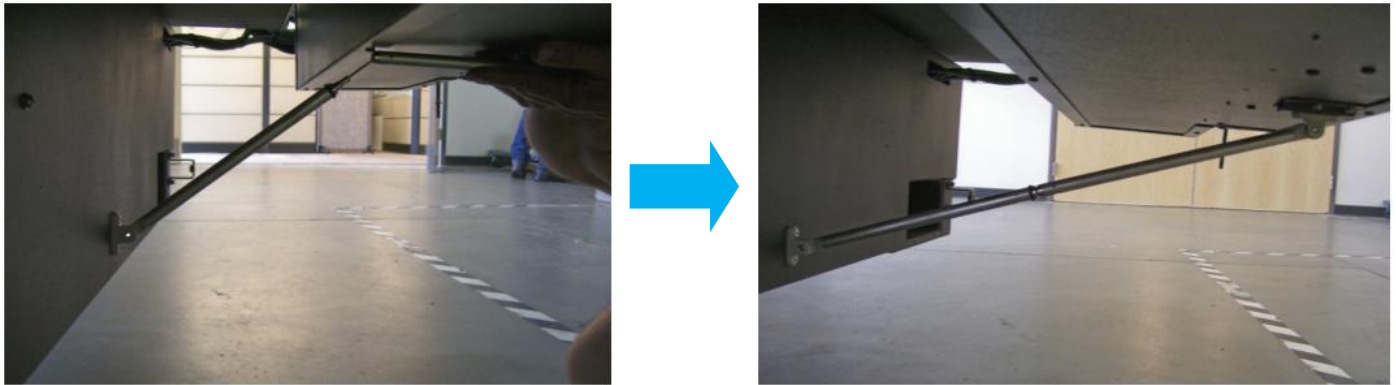


Figure 3 SSCA Support Rod Extend

5. Unlock SSCA and open up SSCA Cover
6. Unplug & remove Defective Relay as shown in Figure 4

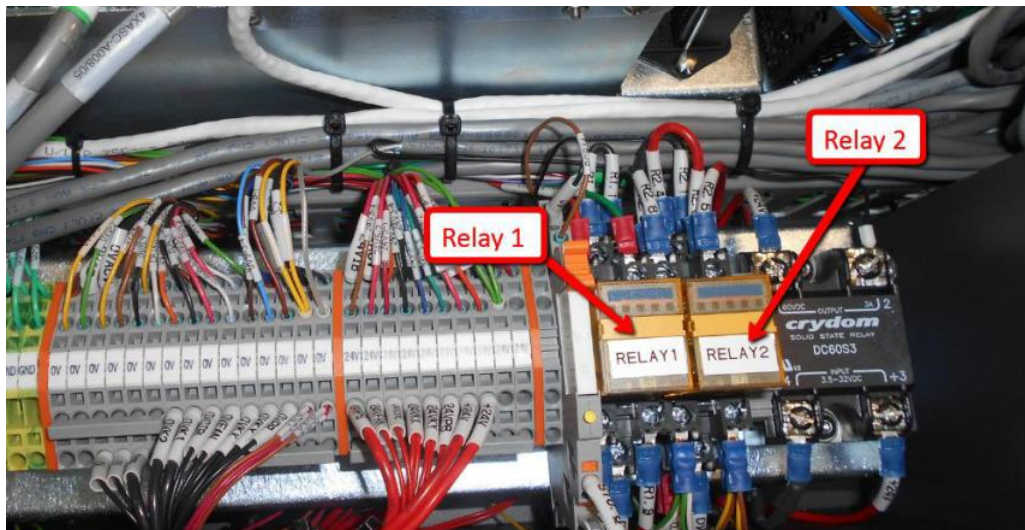


Figure 4 Relay Removal

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Relay replacement

Relay 1 (P/N: 2292-0010): Conveyor Belt On/Off

Relay 2 (P/N: 2292-0010): Conveyor Belt Turning Direction

1. Assemble New Relay.
2. Power on SSCA > Open V810 > Services > Subsystem Status & Control > Digital IO > Initialize

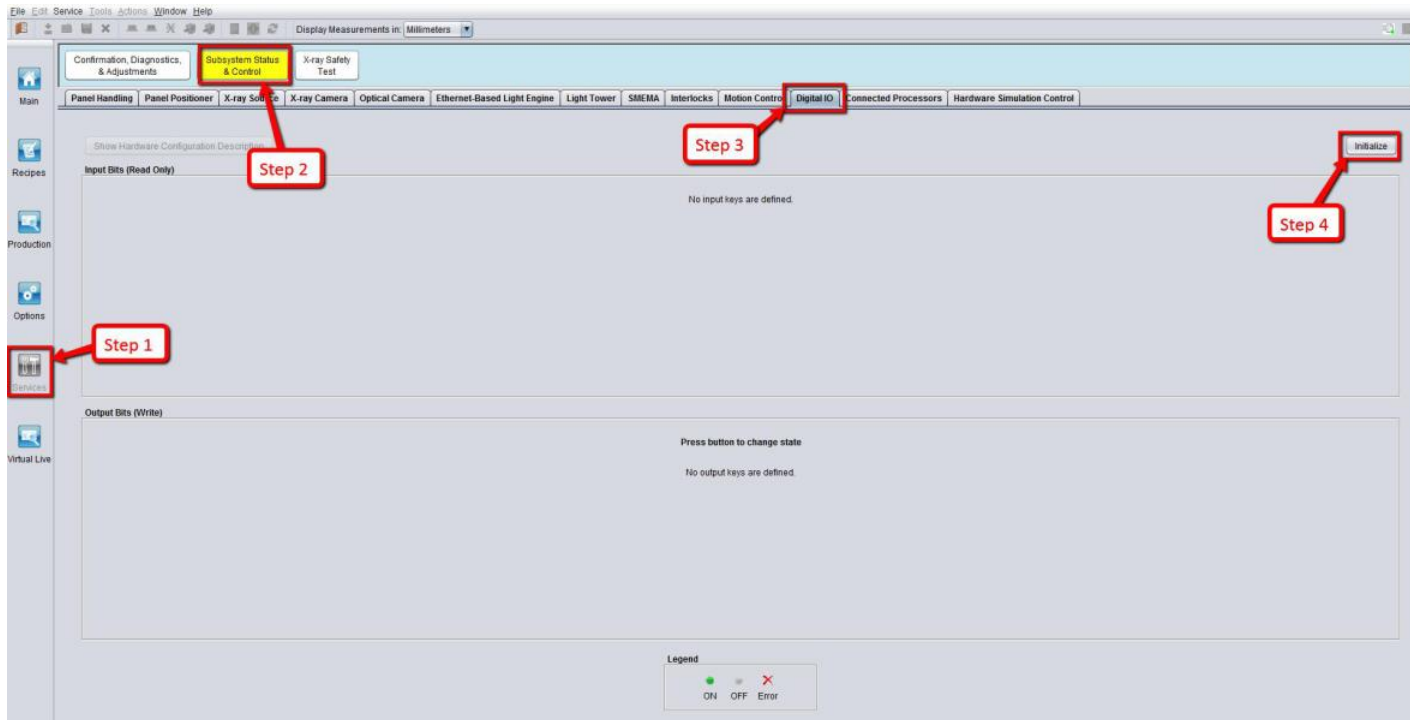


Figure 5 Digital IO Initialize

3. Open Front Access Door
4. Uncheck "Panel Clamps Closed" and "Panel Clamps Opened" will indicate "Green" at Digital IO Input Bit.

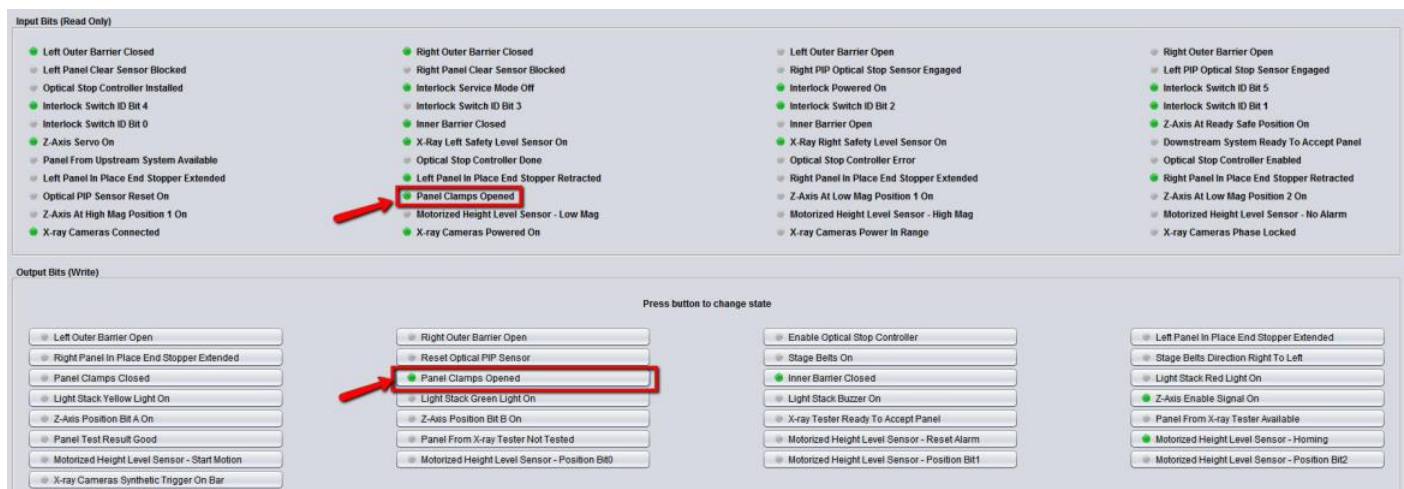
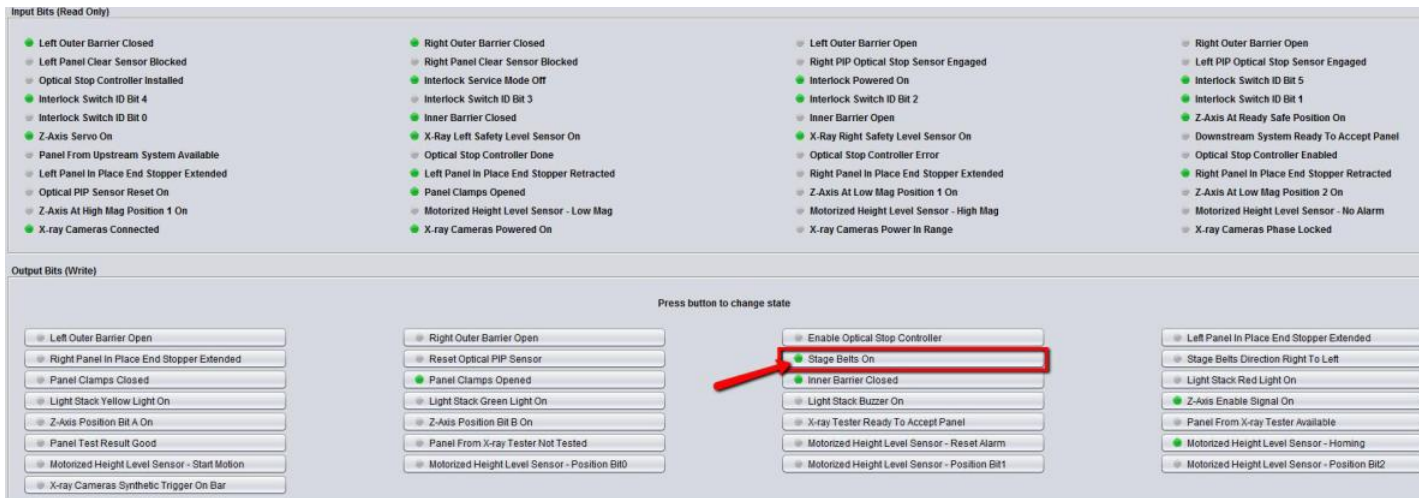


Figure 6 Clamper in "Open" status (Digital IO Input Bit)

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5. Enable “Stage Belts on” as shown in Figure 4. Visual check Belt Running Status at both rail



Input Bits (Read Only)

- Left Outer Barrier Closed
- Left Panel Clear Sensor Blocked
- Optical Stop Controller Installed
- Interlock Switch ID Bit 4
- Interlock Switch ID Bit 0
- Z-Axis Servo On
- Panel From Upstream System Available
- Left Panel In Place End Stopper Extended
- Optical PIP Sensor Reset On
- Z-Axis At High Mag Position 1 On
- X-ray Cameras Connected
- Right Outer Barrier Closed
- Right Panel Clear Sensor Blocked
- Interlock Service Mode Off
- Interlock Switch ID Bit 3
- Inner Barrier Closed
- X-Ray Left Safety Level Sensor On
- Optical Stop Controller Done
- Left Panel In Place End Stopper Retracted
- Panel Clamps Opened
- Motorized Height Level Sensor - Low Mag
- X-ray Cameras Powered On
- Left Outer Barrier Open
- Right PIP Optical Stop Sensor Engaged
- Interlock Powered On
- Interlock Switch ID Bit 2
- Inner Barrier Open
- X-Ray Right Safety Level Sensor On
- Optical Stop Controller Error
- Right Panel In Place End Stopper Extended
- Z-Axis At Low Mag Position 1 On
- Motorized Height Level Sensor - High Mag
- X-ray Cameras Power In Range
- Right Outer Barrier Open
- Left PIP Optical Stop Sensor Engaged
- Interlock Switch ID Bit 5
- Interlock Switch ID Bit 1
- Z-Axis At Ready Safe Position On
- Downstream System Ready To Accept Panel
- Optical Stop Controller Enabled
- Right Panel In Place End Stopper Retracted
- Z-Axis At Low Mag Position 2 On
- Motorized Height Level Sensor - No Alarm
- X-ray Cameras Phase Locked

Output Bits (Write)

Press button to change state

- Left Outer Barrier Open
- Right Panel In Place End Stopper Extended
- Panel Clamps Closed
- Light Stack Yellow Light On
- Z-Axis Position Bit A On
- Panel Test Result Good
- Motorized Height Level Sensor - Start Motion
- X-ray Cameras Synthetic Trigger On Bar
- Right Outer Barrier Open
- Reset Optical PIP Sensor
- Panel Clamps Opened
- Light Stack Green Light On
- Z-Axis Position Bit B On
- Panel From X-ray Tester Not Tested
- Motorized Height Level Sensor - Position Bit0
- Enable Optical Stop Controller
- Stage Belts On**
- Inner Barrier Closed
- Light Stack Buzzer On
- X-ray Tester Ready To Accept Panel
- Motorized Height Level Sensor - Reset Alarm
- Motorized Height Level Sensor - Position Bit1
- Left Panel In Place End Stopper Extended
- Stage Belts Direction Right To Left
- Light Stack Red Light On
- Z-Axis Enable Signal On
- Panel From X-ray Tester Available
- Motorized Height Level Sensor - Homing
- Motorized Height Level Sensor - Position Bit2

Figure 7 Stage Belt in “On” Status

6. Enable “Stage Belts Direction Right To Left” as shown in Figure 5. Visual check Belt Turning Direction.



Input Bits (Read Only)

- Left Outer Barrier Closed
- Left Panel Clear Sensor Blocked
- Optical Stop Controller Installed
- Interlock Switch ID Bit 4
- Interlock Switch ID Bit 0
- Z-Axis Servo On
- Panel From Upstream System Available
- Left Panel In Place End Stopper Extended
- Optical PIP Sensor Reset On
- Z-Axis At High Mag Position 1 On
- X-ray Cameras Connected
- Right Outer Barrier Closed
- Right Panel Clear Sensor Blocked
- Interlock Service Mode Off
- Interlock Switch ID Bit 3
- Inner Barrier Closed
- X-Ray Left Safety Level Sensor On
- Optical Stop Controller Done
- Left Panel In Place End Stopper Retracted
- Panel Clamps Opened
- Motorized Height Level Sensor - Low Mag
- X-ray Cameras Powered On
- Left Outer Barrier Open
- Right PIP Optical Stop Sensor Engaged
- Interlock Powered On
- Interlock Switch ID Bit 2
- Inner Barrier Open
- X-Ray Right Safety Level Sensor On
- Optical Stop Controller Error
- Right Panel In Place End Stopper Extended
- Z-Axis At Low Mag Position 1 On
- Motorized Height Level Sensor - High Mag
- X-ray Cameras Power In Range
- Right Outer Barrier Open
- Left PIP Optical Stop Sensor Engaged
- Interlock Switch ID Bit 5
- Interlock Switch ID Bit 1
- Z-Axis At Ready Safe Position On
- Downstream System Ready To Accept Panel
- Optical Stop Controller Enabled
- Right Panel In Place End Stopper Retracted
- Z-Axis At Low Mag Position 2 On
- Motorized Height Level Sensor - No Alarm
- X-ray Cameras Phase Locked

Output Bits (Write)

Press button to change state

- Left Outer Barrier Open
- Right Panel In Place End Stopper Extended
- Panel Clamps Closed
- Light Stack Yellow Light On
- Z-Axis Position Bit A On
- Panel Test Result Good
- Motorized Height Level Sensor - Start Motion
- X-ray Cameras Synthetic Trigger On Bar
- Right Outer Barrier Open
- Reset Optical PIP Sensor
- Panel Clamps Opened
- Light Stack Green Light On
- Z-Axis Position Bit B On
- Panel From X-ray Tester Not Tested
- Motorized Height Level Sensor - Position Bit0
- Enable Optical Stop Controller
- Stage Belts On
- Inner Barrier Closed
- Light Stack Buzzer On
- X-ray Tester Ready To Accept Panel
- Motorized Height Level Sensor - Reset Alarm
- Motorized Height Level Sensor - Position Bit1
- Left Panel In Place End Stopper Extended
- Stage Belts Direction Right To Left**
- Light Stack Red Light On
- Z-Axis Enable Signal On
- Panel From X-ray Tester Available
- Motorized Height Level Sensor - Homing
- Motorized Height Level Sensor - Position Bit2

Figure 8 Belt Turning Direction (Anti-Clockwise)

7. Close Front Door > Open X-ray source > Run CD&A for 3 loops.

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Crydom SSR Relay

Relay removal

1. Close X-ray Source. Open front door.
2. Power off SSCA

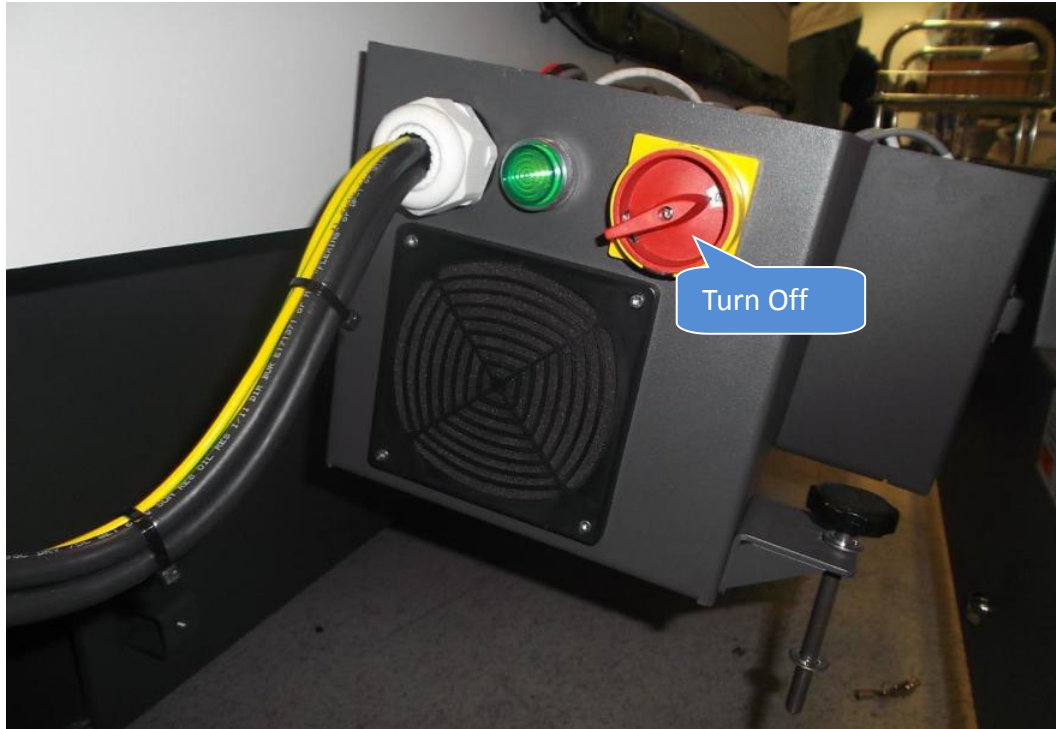


Figure 9 SSCA Power Off

3. Gently loosen Both Anchors as below Figure 2

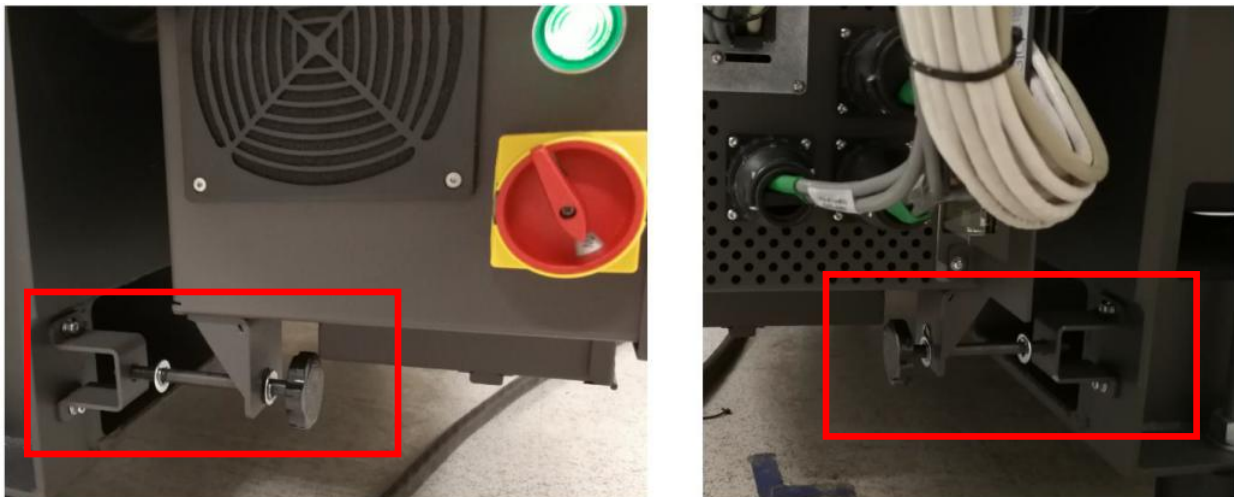


Figure 10 Anchors Release

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4. Extend SSCA Support Rod which locate underneath of SSCA as below Figure 11

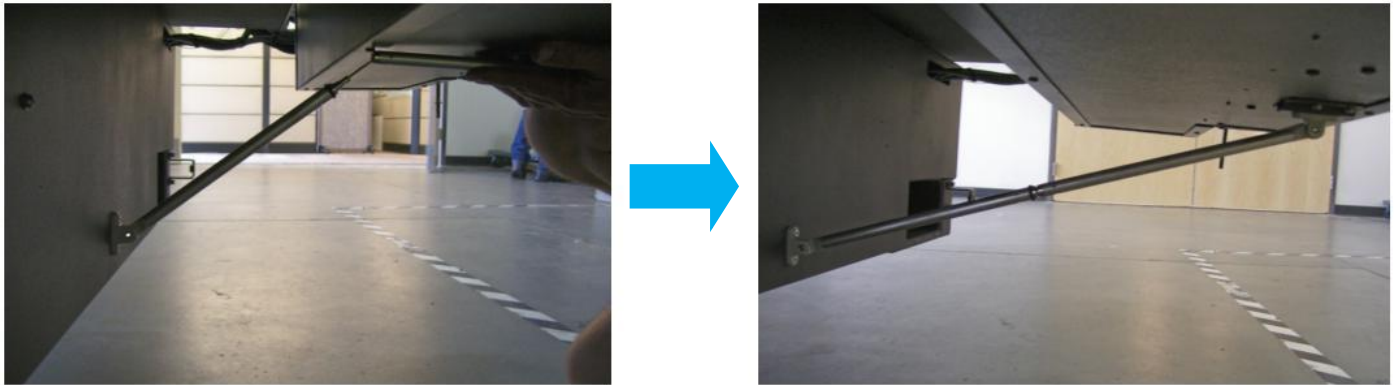


Figure 11 SSCA Support Rod Extend

5. Unlock SSCA and open up SSCA Cover
6. Loosen 4 x Screws and remove 4 x Cables as shown in Figure 12

Remark: Perform Label Marking on 4 x Cable at individual terminal (Terminal 1, 2, 3 & 4).

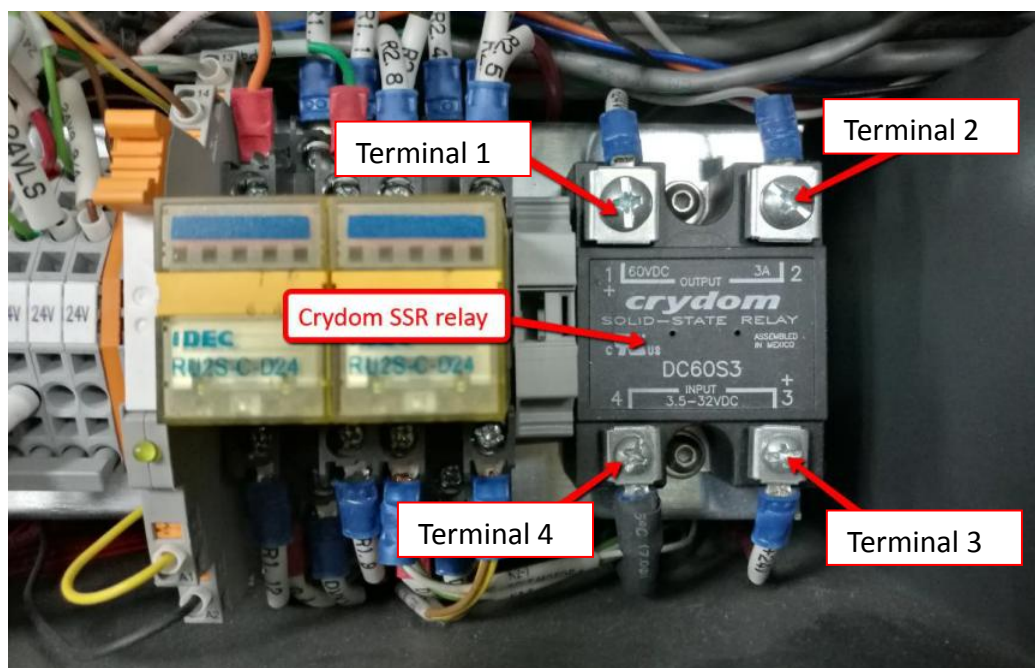


Figure 12 Cables Removal

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- Remove 2 x Screws & Defective Crydom SSR relay as shown in Figure 13

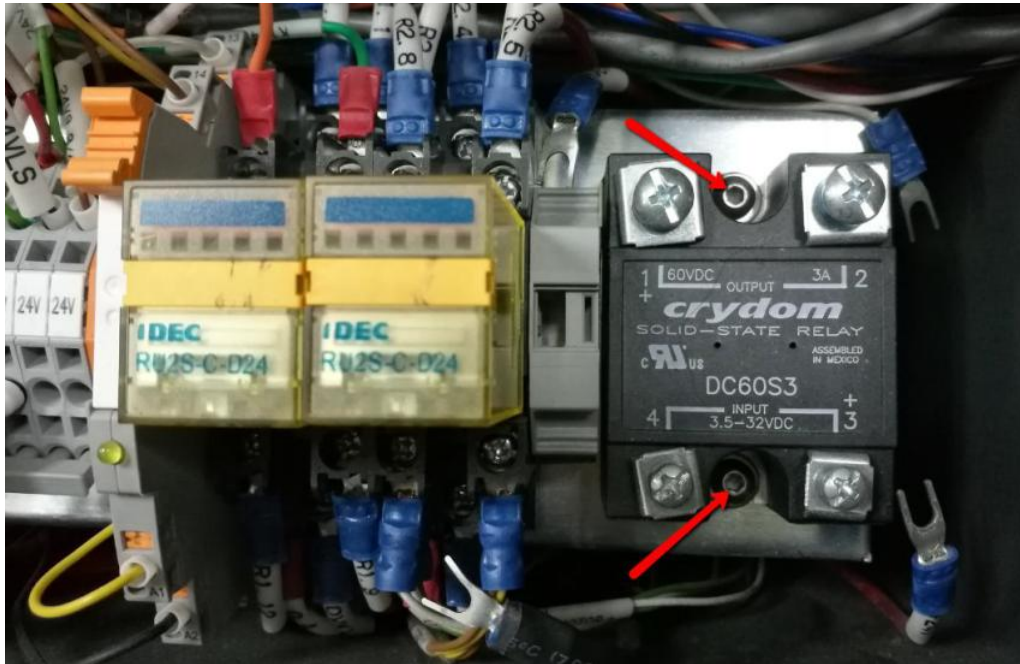


Figure 13 Crydom SSR Relay Removal

Relay replacement

S2EX Crydom SSR relay part ID: 2292-0012

- By refer to below SSR relay , assemble New Relay & tighten 2 x Screws as shown in Figure 14

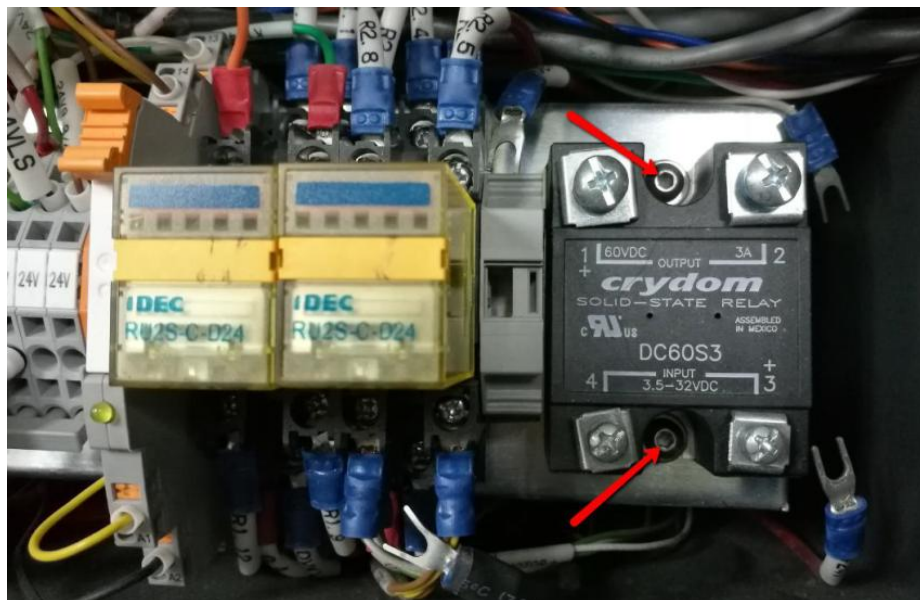


Figure 14 SSR Relay Assembly

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- Refer label marking in Figure 12, gently slot in 4 x Cables into Relay. Tighten 4 x Screws as shown in Figure 15

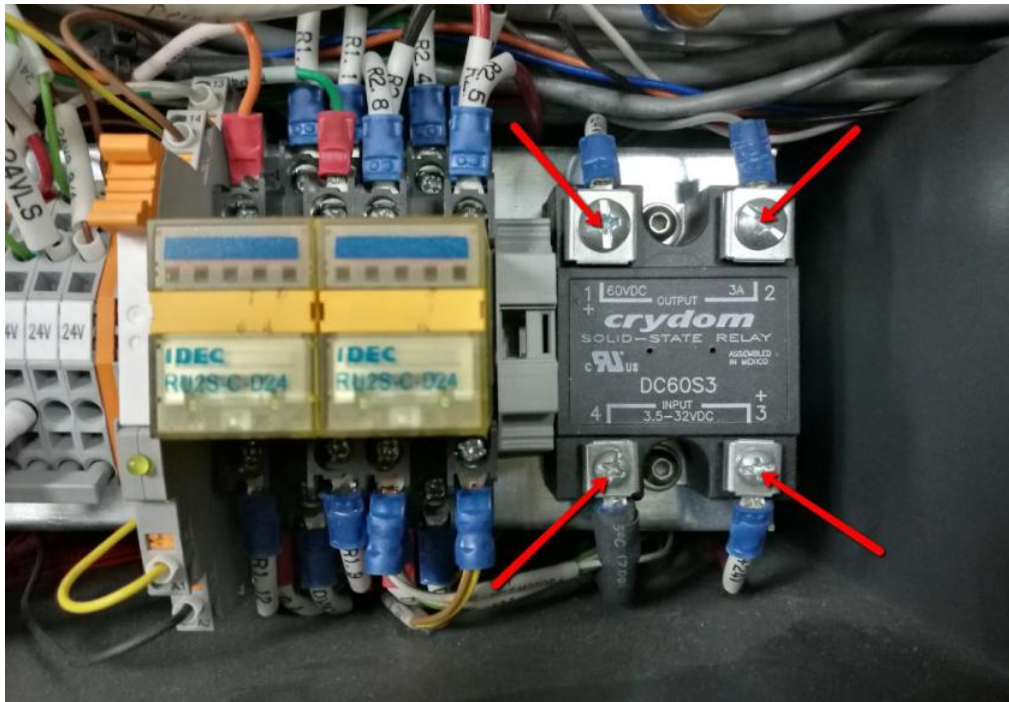


Figure 15 Cables Assembly

- Power on SSCA and close SSCA Cover
- Close all Door Access
- Run 1 loop of CD & A